Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Artificial Intelligence (01CT0616)	**Aim: To learn the Basics of Machine Learning**	
Experiment No: 01	**Date:**	**Enrollment No:92301733021**

Aim: To learn the Basics of Machine Learning

IDE: Google Colab

Theory:

Machine learning is a subset of artificial intelligence in the field of computer science that often uses statistical techniques to give computers the ability to "learn" (i.e., progressively improve performance on a specific task) with data, without being explicitly programmed. In the past decade, machine learning has given us self-driving cars, practical speech recognition, effective web search, and a vastly improved understanding of the human genome.

Machine learning tasks

Machine learning tasks are typically classified into two broad categories, depending on whether there is a learning "signal" or "feedback" available to a learning system:

- **Supervised learning:** The computer is presented with example inputs and their desired outputs, given by a "teacher", and the goal is to learn a general rule that maps inputs to outputs. As special cases, the input signal can be only partially available, or restricted to special feedback:
- **Semi-supervised learning:** the computer is given only an incomplete training signal: a training set with some (often many) of the target outputs missing.
- **Active learning:** the computer can only obtain training labels for a limited set of instances (based on a budget), and also has to optimize its choice of objects to acquire labels for. When used interactively, these can be presented to the user for labeling.
- **Reinforcement learning:** training data (in form of rewards and punishments) is given only as feedback to the program's actions in a dynamic environment, such as driving a vehicle or playing a game against an opponent.
- **Unsupervised learning:** No labels are given to the learning algorithm, leaving it on its own to find structure in its input. Unsupervised learning can be a goal in itself (discovering hidden patterns in data) or a means towards an end (feature learning).

Machine learning applications

In classification, inputs are divided into two or more classes, and the learner must produce a model that assigns unseen inputs to one or more (multi-label classification) of these classes. This is typically tackled in a supervised manner. Spam filtering is an example of classification, where the inputs are email (or other) messages and the classes are "spam" and "not spam". In regression, also a supervised problem, the outputs are continuous rather than discrete.

In clustering, a set of inputs is to be divided into groups. Unlike in classification, the groups are not known beforehand, making this typically an unsupervised task. Density estimation finds the distribution of inputs in

some space. Dimensionality reduction simplifies inputs by mapping them into a lower- dimensional space. Topic modeling is a related problem, where a program is given a list of human language documents and is tasked with finding out which documents cover similar topics.

Machine learning Approaches

Decision tree learning: Decision tree learning uses a decision tree as a predictive model, which maps observations about an item to conclusions about the item's target value. Association rule learning Association rule learning is a method for discovering interesting relations between variables in large databases.

Artificial neural networks

An artificial neural network (ANN) learning algorithm, usually called "neural network" (NN), is a learning algorithm that is vaguely inspired by biological neural networks. Computations are structured in terms of an interconnected group of artificial neurons, processing information using a connectionist approach to computation. Modern neural networks are non-linear statistical data modeling tools. They are usually used to model complex relationships between inputs and outputs, to find patterns in data, or to capture the statistical structure in an unknown joint probability distribution between observed variables.

Deep learning

Falling hardware prices and the development of GPUs for personal use in the last few years have contributed to the development of the concept of deep learning which consists of multiple hidden layers in an artificial neural network. This approach tries to model the way the human brain processes light and sound into vision and hearing. Some successful applications of deep learning are computer vision and speech recognition.

Inductive logic programming

Inductive logic programming (ILP) is an approach to rule learning using logic programming as a uniform representation for input examples, background knowledge, and hypotheses. Given an encoding of the known background knowledge and a set of examples represented as a logical database of facts, an ILP system will derive a hypothesized logic program that entails all positive and no negative examples. Inductive programming is a related field that considers any kind of programming languages for representing hypotheses (and not only logic programming), such as functional programs.

Support vector machines

Support vector machines (SVMs) are a set of related supervised learning methods used for classification and regression. Given a set of training examples, each marked as belonging to one of two categories, an SVM training algorithm builds a model that predicts whether a new example falls into one category or the other.

Clustering

Cluster analysis is the assignment of a set of observations into subsets (called clusters) so that observations within the same cluster are similar according to some pre designated criterion or criteria, while observations drawn from different clusters are dissimilar. Different clustering techniques make different assumptions on the

structure of the data, often defined by some similarity metric and evaluated for example by internal compactness (similarity between members of the same cluster) and separation between different clusters. Other methods are based on estimated density and graph connectivity. Clustering is a method of unsupervised learning, and a common technique for statistical data analysis.

Bayesian networks

A Bayesian network, belief network or directed acyclic graphical model is a probabilistic graphical model that represents a set of random variables and their conditional independencies via a directed acyclic graph (DAG). For example, a Bayesian network could represent the probabilistic relationships between diseases and symptoms. Given symptoms, the network can be used to compute the probabilities of the presence of various diseases. Efficient algorithms exist that perform inference and learning.

Reinforcement learning

Reinforcement learning is concerned with how an agent ought to take actions in an environment so as to maximize some notion of long-term reward. Reinforcement learning algorithms attempt to find a policy that maps states of the world to the actions the agent ought to take in those states. Reinforcement learning differs from the supervised learning problem in that correct input/output pairs are never presented, nor sub-optimal actions explicitly corrected.

Similarity and metric learning

In this problem, the learning machine is given pairs of examples that are considered similar and pairs of less similar objects. It then needs to learn a similarity function (or a distance metric function) that can predict if new objects are similar. It is sometimes used in Recommendation systems.

Genetic algorithms

A genetic algorithm (GA) is a search heuristic that mimics the process of natural selection, and uses methods such as mutation and crossover to generate new genotype in the hope of finding good solutions to a given problem. In machine learning, genetic algorithms found some uses in the 1980s and 1990s. Conversely, machine learning techniques have been used to improve the performance of genetic and evolutionary algorithms.

Rule-based machine learning

Rule-based machine learning is a general term for any machine learning method that identifies, learns, or evolves "rules" to store, manipulate or apply, knowledge. The defining characteristic of a rule-based machine learner is the identification and utilization of a set of relational rules that collectively represent the knowledge captured by the system. This is in contrast to other machine learners that commonly identify a singular model that can be universally applied to any instance in order to make a prediction. Rule-based machine learning approaches include learning classifier systems, association rule learning, and artificial immune systems.

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Feature selection approach

Feature selection is the process of selecting an optimal subset of relevant features for use in model construction. It is assumed the data contains some features that are either redundant or irrelevant, and can thus be removed to reduce calculation cost without incurring much loss of information. Common optimality criteria include accuracy, similarity and information measures.

Pre Lab Exercise:

a. **Write 5 applications, that you encounter daily, where Machine Learning is used?**

   Machine learning is used in apps like YouTube or Netflix for recommendations, Google Maps for traffic prediction, email spam filtering, voice assistants like Alexa, and online shopping suggestions on Amazon. These systems learn from your behavior to improve results.

b. **What do you mean by Supervised Learning?**

   Supervised learning is a type of machine learning where the model is trained using labeled data (input with correct output). It learns to predict results based on examples, like predicting house prices or classifying emails as spam or not.

c. **What do you mean by Unsupervised Learning?**

   Unsupervised learning works with unlabeled data, meaning there are no predefined outputs. The model finds patterns or groups on its own, such as customer segmentation or grouping similar data points.

d. **What do you mean by Reinforcement Learning?**

   Reinforcement learning is where a model learns by interacting with an environment and receiving rewards or penalties. It improves its performance over time, like how a game-playing AI learns to win by trial and error.

Program (Code):

Perform the following tasks:

1. **Load a dataset in your IDE**

Code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
url = "https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv"
df = pd.read_csv(url)

print("Dataset Loaded Successfully!\n")
```

Output:

```
Dataset Loaded Successfully!
```

2. Observe the statistics of all the features

Code:

```
print("Statistical Summary:\n")
print(df.describe())
```

Output:

```
•••  Statistical Summary:

       PassengerId    Survived      Pclass         Age       SibSp  \
count   891.000000  891.000000  891.000000  714.000000  891.000000
mean    446.000000    0.383838    2.308642   29.699118    0.523008
std     257.353842    0.486592    0.836071   14.526497    1.102743
min       1.000000    0.000000    1.000000    0.420000    0.000000
25%     223.500000    0.000000    2.000000   20.125000    0.000000
50%     446.000000    0.000000    3.000000   28.000000    0.000000
75%     668.500000    1.000000    3.000000   38.000000    1.000000
max     891.000000    1.000000    3.000000   80.000000    8.000000

             Parch        Fare
count  891.000000  891.000000
mean     0.381594   32.204208
std      0.806057   49.693429
min      0.000000    0.000000
25%      0.000000    7.910400
50%      0.000000   14.454200
75%      0.000000   31.000000
max      6.000000  512.329200
```

3. Obtain the shape of the dataset

Code:

```
print("\nShape of Dataset:")
print(df.shape)
```

Output:

```
...
    Shape of Dataset:
    (891, 12)
```

4. Separate all the features

Code:

```
X = df.drop("Survived", axis=1)
y = df["Survived"]
print("\nFeatures separated successfully!")
```

Output:

```
...
    Features separated successfully!
```

5. Fill the missing values, if any, using the statistically relevant value

Code:

```
print("\nMissing Values:\n")
print(df.isnull().sum())
```

Output:

```
...
    Missing Values:

    PassengerId      0
    Survived         0
    Pclass           0
    Name             0
    Sex              0
    Age            177
    SibSp            0
    Parch            0
    Ticket           0
    Fare             0
    Cabin          687
    Embarked         2
    dtype: int64
```

Code:

```
# Age -> mean
df['Age'].fillna(df['Age'].mean(), inplace=True)
# Embarked -> mode
df['Embarked'].fillna(df['Embarked'].mode()[0], inplace=True)
# Cabin -> drop (too many missing)
df.drop('Cabin', axis=1, inplace=True)
print("\nMissing values handled!")
```

Output:

```
...
    Missing values handled!
    /tmp/ipykernel_502/3181208361.py:3: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series thro
    The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which w

    For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df


      df['Age'].fillna(df['Age'].mean(), inplace=True)
    /tmp/ipykernel_502/3181208361.py:6: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series thro
    The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which w

    For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df


      df['Embarked'].fillna(df['Embarked'].mode()[0], inplace=True)
```

6. Observe the Box-Plot of each feature

Code:

```
numerical_features = df.select_dtypes(include=np.number).columns

for col in numerical_features:
    plt.figure()
    sns.boxplot(x=df[col])
    plt.title(f"Boxplot of {col}")
    plt.show()
```

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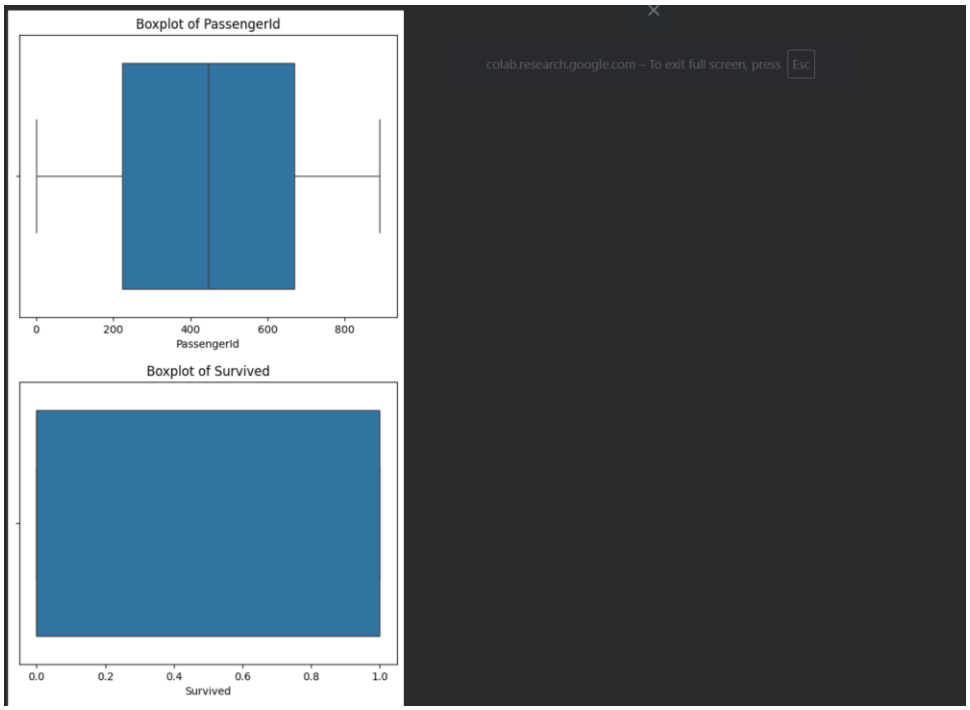
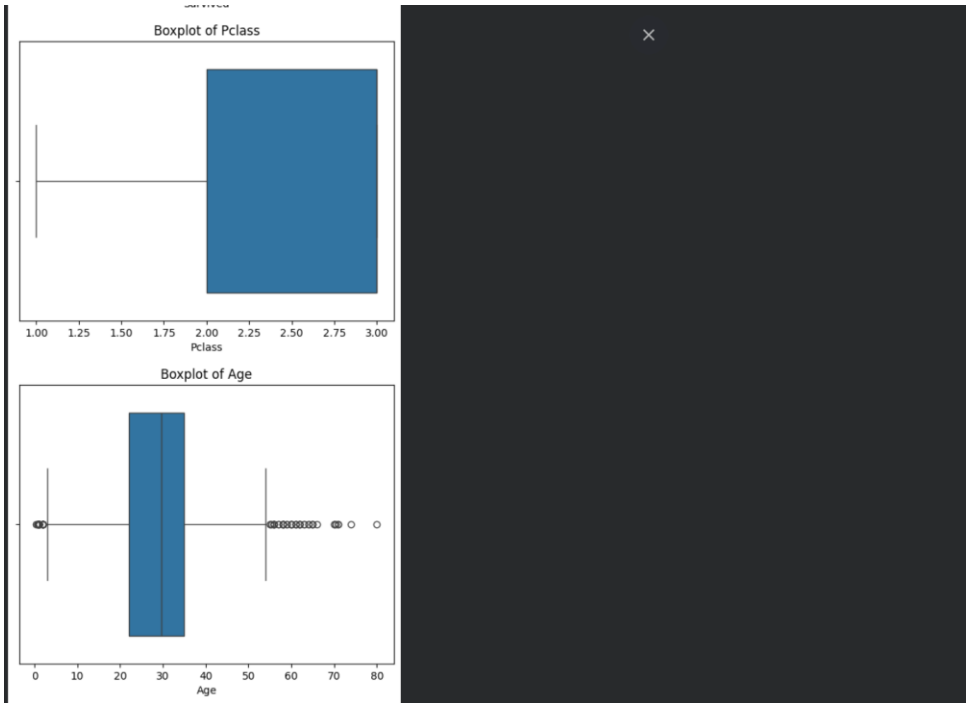
Output:





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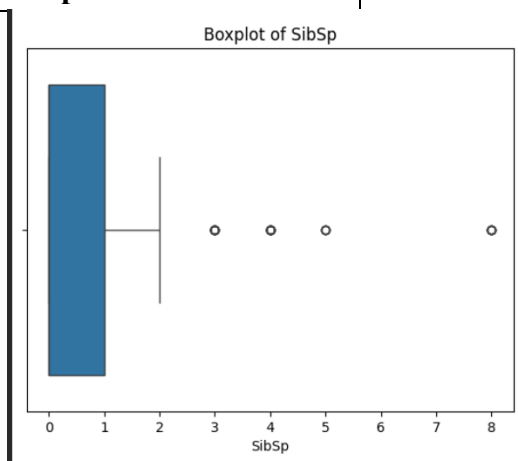

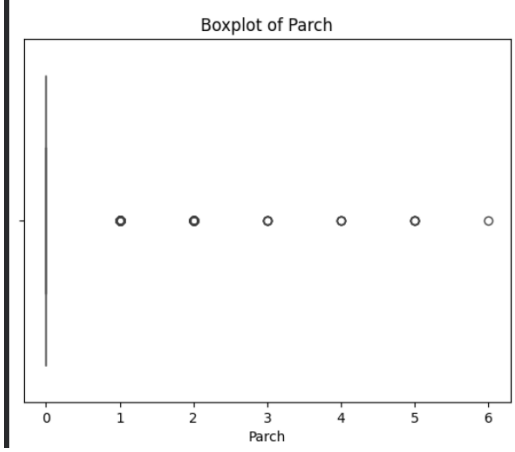

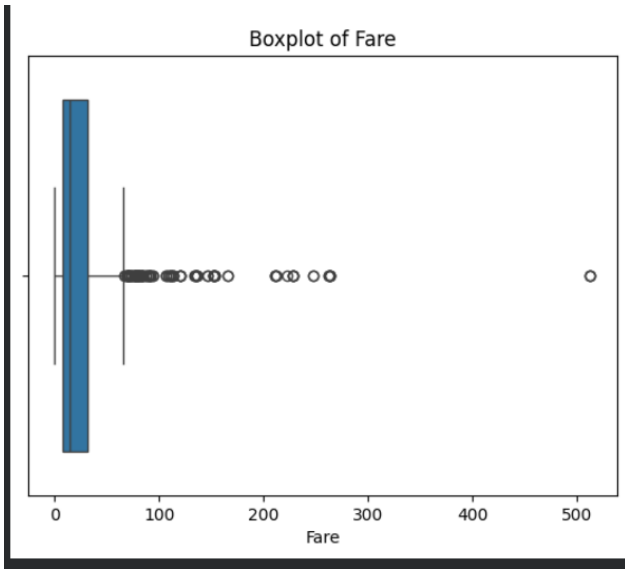

7. Comment on the IQR range and outliers

Code:

```
print("\nIQR and Outliers:\n")
for col in numerical_features:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1
    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR
    outliers = df[(df[col] < lower) | (df[col] > upper)]
    print(f"{col}:")
    print(f"IQR = {IQR}")
    print(f"Outliers = {len(outliers)}\n")
```

Output:

```
IQR and Outliers:

PassengerId:
IQR = 445.0
Outliers = 0

Survived:
IQR = 1.0
Outliers = 0

Pclass:
IQR = 1.0
Outliers = 0

Age:
IQR = 13.0
Outliers = 66

SibSp:
IQR = 1.0
Outliers = 46

Parch:
IQR = 0.0
Outliers = 213

Fare:
IQR = 23.0896
Outliers = 116
```

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Observation and Result Analysis:

a. **Statistics of the dataset**

The dataset shows important statistical measures like mean, median, standard deviation, minimum, and maximum values for numerical features. It helps in understanding the overall distribution and spread of data. For example, features like Age and Fare have noticeable variation, indicating diverse passenger information.

b. **IQR and Outliers**

The IQR (Interquartile Range) represents the spread of the middle 50% of the data. Values lying outside the range $Q1 - 1.5 \times IQR$and $Q3 + 1.5 \times IQR$are considered outliers. In this dataset, features like Fare contain several outliers due to very high values compared to the rest.

Post Lab Exercise:

a. **Which is a better mean of replacing the missing value-mean, median or mode?**

It depends on the data. Mean works well for normally distributed data, median is better when outliers are present, and mode is used for categorical data. In most real cases, median is safer because it is less affected by extreme values.

b. **How IQR range helps to understand the nature of the distribution of the data?**

IQR shows the spread of the middle 50% of the data, helping us understand how values are distributed. It also helps detect outliers and indicates whether the data is tightly packed or widely spread.

c. **Which are the ways to replace the categorical data to the numerical ones?**

Categorical data can be converted using methods like Label Encoding (assigning numbers), One-Hot Encoding (creating binary columns), and Ordinal Encoding (for ordered categories). These methods help machine learning models process non-numeric data easily.